

THE LANGUAGE OF BUSINESS ANALYTICS

SECTION 1 : FOUNDATIONAL CONCEPTS OF ANALYTICS

① DEFINITION

Business Analytics is the broad discipline of using data, statistical methods, and quantitative models to drive business decisions. It covers the full journey from collecting data to applying insights that shape strategy. It always connects findings back to a specific business goal.

REAL WORLD EXAMPLE

Starbucks uses business analytics to decide where to open new stores and which drinks to promote, based on local sales data, foot traffic and customer preference.

SOURCE: Evans, J. R. (2020). Business Analytics: methods, models, and Decisions (3rd ed.). Pearson.

② DEFINITION

Data Analytics is the branch of business analytics that deals with the hands on work of inspecting data sets to uncover patterns, trends and connections. It focuses on the "how" of working with data, whereas business analytics focuses on the "why", how those findings serve business goals. Data analytics does not require a specific business context to exist.

REAL WORLD EXAMPLE

Netflix's data analytics team ^{studied} ~~studying~~ viewing behaviour and found that certain documentaries spike on weekend evenings. That purely analytical finding was then used by the business to guide content purchases.

SOURCE: Provost, F.; & Fawcett, T. (2013). Data Science for Business. O'Reilly media.

③ DEFINITION

Business Intelligence refers to the tools, technologies and practices used for reporting and monitoring business performance. It shows what has happened rather than predicting. BI systems provides dashboards, reports so that managers can see the current state of operations.

REAL WORLD EXAMPLE

A Walmart purchasing manager checks a BI dashboard every morning to see yesterday's sales and stock levels by category, and decides whether urgent restocking is needed.

SOURCE: Chen, H., Chiang, R. H., & Storey, V. C. (2012). Business Intelligence and Analytics: From Big Data to Big Impact. *MIS Quarterly*, 36(4), 1165-1188.

④ DEFINITION

Data-Driven Decision making is a management philosophy where business choices are primarily based on data analysis rather than intuition, experience or seniority alone. It requires decision makers to actively seek relevant data and to adjust their judgement when the evidence points elsewhere.

REAL WORLD EXAMPLE

Amazon's recommendation system decides which products to show on the homepage based on user behaviour data rather than editorial choice which is a data driven strategy that drives roughly 35% of its sales.

SOURCE: Brynjolfsson, E., & McAfee, A. (2014). *The Second machine Age*. W. W. Norton & Company.

⑤ DEFINITION

Big Data refers to datasets so large they exceed the processing capacity of traditional database software. Big data is not an end in itself as

the value lies in extracting insights from it, often using distributed frameworks like Hadoop or Spark.

REAL WORLD EXAMPLE

Facebook processes over 500 TB of new data daily, including posts, likes, video views, and ad clicks, which is used to optimise news feed ranking and ad targeting.

SOURCE : Gondomi, A., & Haider, M. (2015). Beyond the hype : Big data concepts, methods and analytics. *International Journal of Informatic management*, 35(2), 137-144.

⑥ DEFINITION

Descriptive Analytics answers the question "what happened"? by summarising and visualising historical data. It is the most basic level of analytical maturity, usually delivered as reports, dashboards and simple statistics. Descriptive analytics provides the factual foundation for deeper analysis.

REAL WORLD EXAMPLE

An e-commerce company produces a weekly sales report showing revenue, order volume, and return rates by category, giving management a clear picture of last week's performance.

SOURCE : Delen, D., & Demirkan, H. (2013). Data, information and analytics as services. *Decision Support Systems*, 55(1), 359-363.

⑦ DEFINITION

Diagnostic Analytics answers "why did it happen"? by digging deeper into data to identify causes or correlations behind a result. It usually requires finer-grained data and multi dimensional comparison. It is the bridge between knowing what happened and predicting what will happen next.

REAL WORLD EXAMPLE

A SaaS company noticed churn rose by 3% points last month. Diagnostic analysis revealed that lost customers were concentrated in one industry and had below average usage frequently.

SOURCE: Wang, G., Gunasekaran, A., Ngai, E. W., & Papadopoulos, T. (2016). Big data analytics in logistics and supply chain management. *International Journal of Production Economics*, 176, 98-110.

8) DEFINITION

Predictive Analytics answers "what is likely to happen?" using historical data, statistical models and machine learning to estimate future outcome. Its output is probabilistic forecasts rather than certainties. It is widely used in credit scoring, demand forecasting and churn prediction.

REAL WORLD EXAMPLE

Credit card companies use predictive tools/models to judge within seconds whether a transaction is fraudulent based on unusual spending patterns, and decide whether to block it.

SOURCE: Shmueli, G., & Koppius, O. R. (2011). Predictive analytics in information systems research. *MIS Quarterly*, 35(3), 553-571.

9) DEFINITION

Prescriptive Analytics answers "what should we do about it?" by recommending optimal actions on top of predictions. It combines optimisation algorithms, simulation, and decision rules that account for constraints and objectives. It represents the highest level of analytical maturity.

REAL WORLD EXAMPLE

Airlines use prescriptive systems to automatically reassign crew and

gates when flights are delayed, minimising disruption to the wider network.

SOURCE: Bertsimas, D., & Kallus, N. (2020). From predictive to prescriptive analytics. *Management Science*, 66(3), 1025-1044.

⑩ DEFINITION

Data literacy is the ability to read, understand, analyse and communicate with data. It is not just a technical skill, it includes critically assessing data sources, spotting bias and turning insights into clear narratives. In data-driven organisations, it is a baseline capacity for every role.

REALWORLD EXAMPLE

P&G requires all brand managers to complete data literacy training so they can interpret market research and adjust campaigns themselves, rather than relying only on the analytics department.

SOURCE: D'Ignazio, C., & Bhargava, R. (2016). DataBasic: Design principles, tools and activities for data literacy learners. *The Journal of Community Informatics*, 12(3).

Section 2: Data, Measurement & Reporting.

⑪ DEFINITION

Structured Data is organised in a predefined format, usually rows and columns in relational databases, making it easy to query with SQL and analyse with standard tools. It is neat and manageable but limited to predefined field types. Financial records, customer tables and sales logs are typical examples.

REAL WORLD EXAMPLE:

A bank's core system stores every transaction with fixed fields (date, amount, account number, type). This structured data supports real-time balance checks and monthly statements.

SOURCE: microsoft. (2022). Understand types of data and data storage. microsoft learn.

⑫ DEFINITION

Unstructured data has no predefined data model or fixed format. It includes text, images, audio, video and social media content. It usually requires specialised tools such as natural language processing for text or computer vision for images. Its share of total enterprise data is rising rapidly.

REAL WORLD EXAMPLE:

An insurance company analyses call centre audio recordings (unstructured audio) to detect customer sentiment and common complaint themes, then improves agent training.

SOURCE: Gandomi, A., & Haider, M. (2015). Beyond the hype: Big data concepts, methods and analytics. *International Journal of Information management*, 35(2), 137-144.

⑬ DEFINITION

A Data Warehouse is a centralised storage system built specifically for analysis and reporting, integrating data from multiple source systems. It typically uses ETL (Extract - Transform - Load) processes to convert operational data into an analysis-ready format. Its core value is providing a "single source of truth" across departments.

REAL WORLD EXAMPLE

A retail chain loads data from 500 store POS systems, its e-commerce platform, and inventory management into one warehouse, giving head office an integrated omnichannel sales view.

SOURCE: Inmon, W. H. (2005). Building the Data Warehouse (4th ed.). Wiley.

⑭ DEFINITION

Data mining is the process of automatically or semi-automatically discovering previously unknown patterns, associations, and rules in large datasets. It uses statistic, machine learning, and database techniques to uncover hidden insights with business value. It is often described as "digging for gold" in data.

REAL WORLD EXAMPLE

A supermarket's data mining revealed that customers buying nappies often also bought beer (the classic "beer and nappies" case), so the two products were placed near each other to boost cross-selling.

SOURCE: Han, J., Pei, J., & Tong, H. (2022). Data Mining: Concepts and Techniques (4th ed.).

⑮ DEFINITION

Data visualisation is the practice of turning data into charts, graphs and interactive interfaces so complex patterns become intuitive. Good visualisation is not just attractive, it communicates accurately and guides correct interpretation. Common forms include bar charts, line graphs, scatter plots and heatmaps.

REAL WORLD EXAMPLE

John's Hopkins university's COVID-19 dashboard used maps and trend

charts to show global case distribution, becoming a primary information source for the public and policy makers.

SOURCE: Few, S. (2012). *Show Me the Numbers: Designing Tables and Graphs to Enlighten* (2nd ed.). Analytics Press.

(16) DEFINITION

A Dashboard is a visual interface that consolidates key metrics and performance data on one screen for real-time or periodic monitoring. It usually combines multiple data sources and presents them as charts, numbers and indicators, designed so key information is visible "at a glance". Unlike static reports, dashboards are interactive and often update live.

REAL WORLD EXAMPLE

Uber's operations team uses a live dashboard to monitor supply-demand match rates, average wait times, and driver availability by city, triggering incentives whenever imbalance appears.

SOURCE: Few, S. (2013). *Information Dashboard Design* (2nd ed.). Analytics Press.

(17) DEFINITION

A Metric is any quantifiable and trackable number that describes some aspects of business activity. Metrics are the basic building blocks of all performance measurement, but not every metric is a Key Performance Indicator. A metric can be an absolute number (monthly sales), a ratio (conversion rate), or an index (satisfaction score).

REAL WORLD EXAMPLE

An e-commerce site tracks "page load time" as a technical metric for site performance, and "cart abandonment rate" as a business metric

for user behaviour.

SOURCE: Parmenter, D. (2015). Key Performance Indicators (3rd ed.). Wiley

18 DEFINITION

A Key Performance Indicator is a selected metric directly tied to an organisation's strategic objectives, used to measure the health of critical business activities. The key difference from an ordinary metric is the word "key"; if this number underperforms, the organisation must act immediately. A good KPI is actionable: managers know what to do when it fails.

REAL WORLD EXAMPLE

For a SaaS company, monthly recurring revenue (MRR) is a core KPI because its growth directly ^{reflects} ~~affects~~ business health, whereas website visits may be ~~the~~ only an ordinary metric.

SOURCE: Parmenter, D. (2015). Key Performance Indicators (3rd ed.). Wiley.; SAP. (2024). Using Key Performance Indicators.

19 DEFINITION

A Benchmark is a reference standard or point of comparison used to evaluate performance. It can be internal (historical data, other departments) or external (industry averages, competitor data). Its value is turning an isolated number into a meaningful judgement - knowing whether something is "good" or "bad" requires a reference point.

REAL WORLD EXAMPLE

A bank's customer satisfaction score of 78 looks meaningless alone, but compared with an industry benchmark of 72, management confirms the bank is outperforming peers.

SOURCE: ServiceNow. (2025), What is a key performance indicator (KPI)?

20 DEFINITION

Segmentation is the process of dividing customers or markets into smaller groups with similar characteristics, so different strategies can be applied to each. Dimensions can include demographics (age, income), behaviour (purchase frequency), psychology (values), or geography. Effective Segmentation requires clear differences between groups and similarity within them.

REAL WORLD EXAMPLE

Spotify segments users by listening behaviour into groups like "fitness fans", "commuters", and "party people", then recommends different playlist styles to each.

SOURCE: Kotler, P., & Keller, K. L. (2016), marketing management (15th ed.). Pearson.

Section 3: Customer & Sales KPI's

21 DEFINITION

Conversion rate is the percentage of visitors who complete a target action. The action varies by business: a purchase for e-commerce, a trial sign up for SaaS, a lead form submission for B2B. It is a core measure of marketing and user experience efficiency, affected by traffic quality, page design, and friction in the conversion path.

REAL WORLD EXAMPLE

An e-commerce site receives 100,000 monthly visitors and 2,500 purchases, giving a 2.5% conversion rate. With an industry average of 2%, the site is slightly above benchmark but still has room to optimise.

SOURCE: Google. (2024). Google Analytics Help: Conversion Rate.

(22) DEFINITION

Customer Acquisition Cost (CAC) is the average cost of acquiring one new customer, calculated by dividing total acquisition spend (marketing and sales) in a period by the number of new customers gained. It is a core measure of acquisition efficiency and is used together with Customer Lifetime Value (CLV) to judge whether a business model is sustainable.

REAL WORLD EXPERIENCE

A DTC brand spends \$500,000 on marketing in a quarter and gains 5000 new customers, giving a CAC of \$100. If the average order value is only \$80 with low repeat purchase, the CAC is too high and the strategy needs adjustment.

SOURCE: Investopedia. (2024). Customer Acquisition Cost (CAC).

(23) DEFINITION

Customer Lifetime Value (CLV/LTV) is the total net profit a business expects to earn from one customer over the entire relationship. It accounts for first purchase, repeat purchases and retention over time. Comparing CLV with CAC is the core framework for unit economics, if CLV does not significantly exceed CAC, growth is simply burning money.

REAL WORLD EXAMPLE

A subscription software costs \$50 per month, retains customers for 30 months on average, and has an 80% gross margin, giving a CLV of about \$1,200. With a CAC of \$300, the CLV:CAC ratio is 4:1, a healthy range.

SOURCE: Kumar, V., & Reinartz, W. (2018). Customer Relationship management (3rd ed.). Springer.

24) DEFINITION

Churn Rate is the percentage of customers who stop using a product or cancel a subscription in a given period. It is one of the most important inverse indicators in subscription business models—high churn means new customers must be consistently acquired just to maintain revenue, pushing acquisition costs up.

EXAMPLE REAL WORLD DEFINITION

A streaming platform starts the month with 1 million subscribers and loses 30 000 cancellations, giving a 3% monthly churn rate. Annualised at roughly 36%, more than a third of users would leave within a year.

SOURCE: Baremetrics. (2026). What is the LTV to CAC Ratio?

25) DEFINITION

Customer Retention Rate is the percentage of customers still using the product or service at the end of a period compared with the start. It is the complement of churn. High retention means customers are satisfied and continuously receiving value, forming the basis of long-term profitability. It is usually measured monthly, quarterly or annually.

REAL WORLD EXAMPLE

An enterprise software company starts the year with 500 business customers and retain 460 by year-end, giving a 92% annual retention rate, above the SaaS industry's excellent standard of 90%.

SOURCE: Reichheld, F. F. (2026). The Ultimate Question. Harvard Business School Press.

26) DEFINITION

Net Promoter Score (NPS) measures customer loyalty and willingness to recommend, based on the question "How likely are you to recommend us to a friend or colleague?" (0-10). Scores of 9-10 are Promoters, 7-8 are Passives and 0-6 are Detractors. $NPS = \% \text{ Promoters} - \% \text{ Detractors}$, ranging from -100 to +100.

REAL WORLD EXAMPLE

Apple's NPS has long stayed around 70, reflecting a highly loyal and vocal user base. The industry average typically falls between 30 and 40.

SOURCE: Reicheld, F. F. (2003). The One Number You Need to Grow. Harvard Business Review, 81(12), 46-54.

27) DEFINITION

Average Order Value (AOV) is the average amount spent per transaction, calculated as total sales divided by number of orders. It is a core metric in e-commerce and retail. Strategies to raise AOV include bundling, threshold discount and upselling. Changes in AOV can reflect pricing, product mix or promotion effectiveness.

REAL WORLD EXAMPLE

A fashion e-commerce site has \$1,000,000 monthly sales and 20,000 orders, giving an AOV of \$50. After launching a "free shipping over \$75" promotion, AOV rises to \$62 and sales grow 15%.

SOURCE: Shopify. (2024). Average Order Value: What It Is and How to Improve It.

28) DEFINITION

The Sales Funnel describes the stages a potential customer passes through from first brand contact to final purchase, typically Awareness, Interest, Consideration, Intent and Purchase. The funnel shape reflects declining conversion at each stage. Analysing the funnel helps identify bottlenecks in the process.

REAL WORLD EXAMPLE

A B2B company found conversion from Consideration to Intent was extremely low. Deeper analysis showed the sales team was too slow to follow up, after fixing this, overall conversion ~~rate~~^{rose} 40%.

SOURCE: HubSpot. (2024). Sales Funnel: What It Is and How to Build One.

Section 4: Marketing & Campaign Analytics

29) DEFINITION

A marketing campaign is an organised, goal-driven marketing effort, usually built around a specific product launch, brand push, or sales target, with a defined timeframe, target audience, channel mix and budget. A campaign can run across multiple channels (social media, email, search ads) in a coordinated way.

REAL WORLD EXAMPLE

Nike's "Just Do It" campaign during the Olympics ran TV ads, social media hashtags, and athlete endorsements simultaneously to lift brand awareness and sportswear sales.

SOURCE: Kotler, P., & Keller, K.L. (2016). marketing management (15th ed), Pearson.

30) DEFINITION

Post - Campaign Analysis is the systematic evaluation of a marketing campaign after it ends, to measure whether goals were met, what worked and what should improve. It usually compares key metrics before and after the campaign (sales, brand search volume, conversion rate) to estimate incremental impact. It informs future marketing decisions.

REAL WORLD EXAMPLE

After a summer promotion, a beverage brand's post-campaign analysis found social media ads drove more incremental sales than TV ads, leading to a budget shift towards digital next quarter.

SOURCE : HubSpot. (2024). How to Run a Post-Campaign Analysis.

31) DEFINITION

Incremental Revenue is the extra revenue directly caused by a marketing campaign - sales that would not have happened without it. It differs from total campaign revenue, which includes sales that would have ^{occurred} ~~occurred~~ naturally. measuring incremental revenue usually requires a control group or attribution modelling.

REAL WORLD EXAMPLE

An e-commerce brand send a \$10 coupon to non-buyers and generates \$50 000 in extra sales; the same coupon sent to existing customers generates \$30 000, most of which would have happened anyway - only the former is true incremental revenue.

SOURCE : Farris, P.W., Bendle, N.T., Pfeifer, P.E., & Reibstein, D.J. (2015) marketing metrics (3rd ed.). Pearson.

32) DEFINITION

Return on Investment is the classic measure of investment efficiency, calculated as $(\text{Gain} - \text{Cost}) / \text{Cost} \times 100\%$. In marketing, ROI measures the net return on marketing spend. ROI can be financial or intangible (brand, awareness), though the latter is harder to quantify.

REAL WORLD EXAMPLE

A brand spends \$200,000 on a campaign that generates \$500,000 incremental revenue. After \$200,000 in product/services costs, net gain is \$100,000, giving an ROI of 50%.

SOURCE: Investopedia. (2024). Return on Investment (ROI).

33) DEFINITION

Return on Ad Spend (ROAS) is the ratio of ad revenue to ad spend, calculated as revenue from ads divided by ad spend. Key difference from ROI: ROAS looks only at ad revenue and does not deduct product or service costs, ~~whereas~~ ^{whereas} ROI is a net profit concept. ROAS = 5 means every \$1 of ad spend returns \$5 in revenue.

REAL WORLD EXAMPLE

An e-commerce ad campaign spends \$20,000 and generates \$100,000 in direct sales, giving a ROAS of 5:1. With a 40% gross margin, gross profit is \$40,000; after \$20,000 ad spend, net profit is \$20,000.

SOURCE: Google. (2024). Google Ads Help: Return on Ad Spend (ROAS).

34) DEFINITION

Click-Through Rate (CTR) is the proportion of users who see an ad or link and actually click it, calculated as clicks divided by impressions. CTR measures how attractive the ad copy, image or headline is. High CTR means

the content caught the target audience's attention, but high CTR does not guarantee high conversion - the landing page experience matters too.

REAL WORLD EXAMPLE

An email campaign sends 10 000 emails and gets 500 clicks, giving a 5% CTR. The industry average is around 2-3%, suggesting the subject line and content were effective.

SOURCE : Google . (2024). Google Analytics Help : Click-Through Rate.

35) DEFINITION

Bounce Rate is the percentage of visitors who leave after viewing only one page. A high bounce rate may mean the landing page does not match user expectations, loads slowly, or has low-quality content. However, it must be interpreted by page type - blog posts naturally have higher bounce rates, while a high bounce rate on a product page is a warning sign.

REAL WORLD EXAMPLE

A new site found one "healthy eating" article had an 85% bounce rate. Analysis showed page load time exceeded 5 seconds, after optimisation, bounce rate fell to 60%.

SOURCE : Google . (2024). Google Analytics Help : Bounce Rate.

36) DEFINITION

A/B Testing is a controlled experiment where an audience is randomly split into two groups shown two versions (A and B), and key metrics (conversion rate, CTR) are compared to see which performs better. The core principle is changing only one variable so differences can be attributed to it. It is a foundational tool for data-driven marketing optimisation.

REAL WORLD EXAMPLE

An e-commerce site tested green vs orange "Add to cart" buttons. The orange version lifted conversion by 12%, and the site adopted orange site-wide.

SOURCE 8 Kohavi, R., Tang, D., & Xu, Y. (2020). Trustworthy Online Controlled Experiments. Cambridge University Press.

37 DEFINITION

Lead Generation is a process of identifying and attracting potential customers with the goal of capturing contact details or behavioural data from people interested in a product or service. Leads can come from content downloads, form submissions, trial signups or event participation. Lead quality matters more than quantity - low quality leads waste sales resources.

REAL WORLD EXAMPLE

A B2B software company requires visitors to submit emails and work details to download an industry whitepaper, generating about 800 leads per month, of which roughly 15% become sales opportunities.

SOURCE 8 HubSpot. (2024). What is Lead Generation?

38 DEFINITION

Marketing Attribution is the process of determining which marketing channel or touchpoint deserves credit for a specific conversion (purchase, Signup). Attribution models range from simple "last click" to complex multi-touch data-driven models. The core challenge is that customers usually pass through multiple touchpoints before converting, so credit must be allocated fairly.

REAL WORLD EXAMPLE

A user sees a Facebook ad, clicks a Google search ad three days later, and finally purchases via an email link. Under "last click" attribution, all credit goes to email; under "linear" attribution, the three channels share credit equally.

SOURCE: Google. (2024). Google Analytics Help: Attribution Models.

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